

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2007

MATHEMATICS

READ THESE INSTRUCTIONS CAREFULLY

Time: 2 hours

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 questions.
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer. Mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only ordinary pencil.
5. Make sure that you have written on the answer sheet:

**YOUR INDEX NUMBER**

**YOUR NAME**

**NAME OF YOUR SCHOOL**

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number (i.e. School Code and three figured Candidate's Number) in the grid near the top of the answer sheet.
7. Do not make any mark outside the box.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the questions 1-50 four answers are given. The answers are lettered A, B, C and D. In each only **ONE** of the answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter chosen is written.

**Example**

**In the question booklet:**

11. What is the value of  $6(24 - 18) + 6 \times 4$ ?

6

- A. 30
- B. 25
- C. 10
- D. 28

The correct answer is C(10)

On the answer sheet:

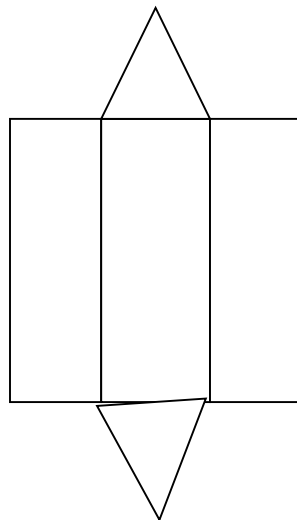
1. [A] [B] [C] [D]    11. [A] [B] [C] [D]    21 [A] [B] [C] [D]    31 [A] [B] [C] [D]

In the set, the box with the letter D printed in it is marked.

12. Your dark line **MUST** be within the box.
  13. For each question **ONLY ONE** box is to be marked in each set of four boxes.
- The Kenya Nation Examination Council, 2007

1. Which one of the following is 1, 100,000.001 written in words?
  - A. One million one hundred thousand and one hundredth.
  - B. One million one thousand and one thousandth.
  - C. One million one hundred thousand one tenth.
  - D. One million one hundred thousand and one thousandth
2. What is the place value of the digit 7 obtained after working out  $0.2016 \div 28$ ?
  - A. Tenths
  - B. Hundredths
  - C. Thousandths
  - D. Ten thousandths
3. What is the 6.0947 rounded off correct to two decimal places?
  - A. 6.15
  - B. 6.10
  - C. 6.095
  - D. 6.09
4. What is the value of  $3\frac{2}{5} - 2\frac{1}{4} \times 1\frac{1}{3} + 1\frac{1}{6}$ ?
  - A.  $2\frac{32}{35}$
  - B.  $1\frac{86}{105}$
  - C.  $24/35$
  - D.  $4/35$
5. What is the volume of cylindrical tank of diameter 5m and a height 6m? (Take  $\pi = 22/7$ ).
  - A.  $471\frac{3}{7}\text{m}^3$
  - B.  $133\frac{4}{7}\text{m}^3$
  - C.  $117\frac{6}{7}\text{m}^3$
  - D.  $94\frac{2}{7}\text{m}^3$
6. Auma and Maina shared the profit from the sale of fish in the ratio 3:4. If the profit was sh. 8,400, how much money did Auma get?
  - A. Sh. 3,600
  - B. Sh. 4 800
  - C. Sh. 6 300
  - D. Sh. 11, 200
7. Mariam bought the following items:  
**3kg flour @ 55**  
**2kg packets of sugar for sh.115**  
**2 bar of soap @ sh. 42**  
**3litres of milk @ sh.25**  
**1  $\frac{1}{2}$  litres of kerosene @ sh. 45**  
If she gave the shopkeeper a sh. 100 note, how much balance did she receive?
  - A. Sh. 378.50
  - B. Sh. 493.50
  - C. Sh. 718.00

- D. Sh. 506.50
8. A map is drawn to a scale of 1: 20 000. What is the distance, in kilometers, of a road which is 25 cm on the map?
- A. 5  
B. 50  
C. 500  
D. 5,000
9. By how much is L.C.M of 40 and 60 more than their GCD?
- A. 20  
B. 100  
C. 120  
D. 140
10. What is the value of  $\frac{1.5 + 0.9 \text{ of } 2.5 - 1.42}{0.4}$ ?
- A. 5.48  
B. 5.825  
C. 5.88  
D. 10.85
11. Below is the net of a solid. The shaded parts are the flaps.

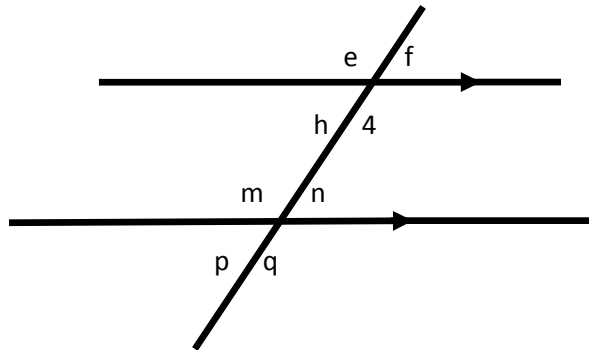


Which one of the following solids can be formed from the net?

- A. Rectangular prism

- B. Triangular pyramid
  - C. Rectangular pyramid
  - D. Triangular prism.
12. Which one of the following quadrilaterals has only one pair of parallel lines?
- A. Rhombus
  - B. Trapezium
  - C. Parallelogram
  - D. Rectangle
12. A fruit vendor prepared juice which filled eight 3-litres container. He later put the juice in 2-deciliter bottles for sale. How many such bottles of juice did he get?
- A. 12
  - B. 120
  - C. 1,200
  - D. 12,000
13. A lawn is in the form of a rectangle and two semicircle. The rectangle is 22m long and 6m wide. The widths of the rectangle are also the diameters of the semicircle. What is the area of the lawn in  $m^2$ ? (Take  $\pi = 22/7$ )
- A. 12
  - B. 120
  - C. 1,200
  - D. 12,000
14. A lawn is in form of a rectangle and two semicircles. The rectangle is 22m long and 6m wide. The widths of the rectangle are also the diameters of the semicircles. What is the area of the lawn in  $m^2$ ?
- A.  $245\frac{1}{2}$
  - B.  $188\frac{1}{2}$
  - C.  $160\frac{1}{2}$
  - D.  $28\frac{1}{2}$
15. Ali and Maingi sat for five subjects in an examination. The total marks for Maingi were 15 less than those of Ali. The mean mark for Ali was 65. What was the mean mark for Maingi?
- A. 50
  - B. 62
  - C. 68
  - D. 310
16. What is the next number in the pattern 16,44,82,130,-----
- A. 188
  - B. 212
  - C. 178
  - D. 140

17. In a certain village the total number of women and children was 34,603. The number of women and men was 18,623. If the number of men was 5,984, what was the number of children in that village?
- A. 12,639  
B. 15,980  
C. 28,619  
D. 21,964
18. The figure below shows angle formed by a pair of parallel lines and a transversal.



In which group below, are each of the angles equal to  $n$ ?

- A.  $p, g, f$   
B.  $p, m, f$   
C.  $p, q, f$   
D.  $p, h, f$

19. What is the value of  $\sqrt{4.2849}$

- A. 0.207  
B. 2.07  
C. 2.7  
D. 207

20. In a certain school, the fractions of boys in classes 5, 6, and 8 are as follows: Class 5 is  $\frac{12}{16}$ , Class 6 is  $\frac{13}{26}$ , Class 7 is  $\frac{10}{25}$  and class 8 is  $\frac{10}{18}$ .

Which class has the least number of boys if all the classes have the same number of pupils?

- A. Class 5  
B. Class 6  
C. Class 7  
D. Class 8
21. What is the value of  $x$  in the equation

$$\frac{2x-3}{3} + 2x = 6?$$

- A.  $1\frac{1}{8}$   
B.  $5\frac{7}{8}$   
C.  $1\frac{7}{8}$

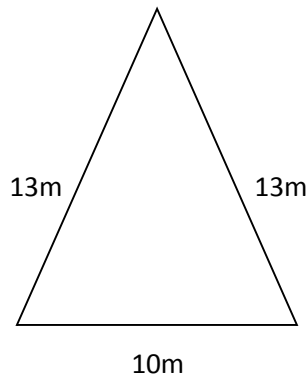
D.  $2\frac{5}{8}$

22. The table below shows the number of the vehicles that passed near a school in one week. The average number of vehicle per day was 116. The number of vehicles that passed near the school on Friday is not shown.

| MON | TUE | WED | THUR | FRY   | SAT | SUN |
|-----|-----|-----|------|-------|-----|-----|
| 125 | 75  | 112 | 100  | _____ | 148 | 112 |

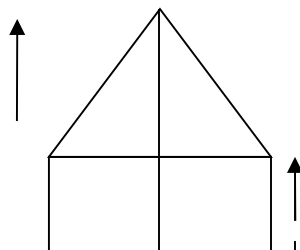
How many more vehicles passed near the school on Friday than on Tuesday?

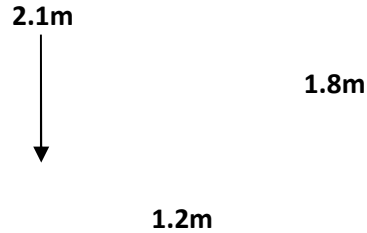
- A. 595  
 B. 215  
 C. 140  
 D. 65
23. A small aircraft took 6 hours 30 minutes to travel from Pretoria to Mombasa. It reached Mombasa at 04 00h on Wednesday. At what time and day did it depart from Pretoria?
- A. 09 30h on Wednesday  
 B. 09 30h on Tuesday  
 C. 21 30h on Wednesday  
 D. 21 30h on Tuesday
24. The diagram below represents a flower garden.



What is the area of the flower garden in  $\text{m}^2$ ?

- A. 36  
 B. 60  
 C. 65  
 D. 120
25. The diagram below represents a door which was painted on the outside.





What was the area o painted, square metres?

- A. 1.17
  - B. 2.16
  - C. 2.34
  - D. 2.52
26. What one of the following groups of measurements represents length of sides of a right angled triangle?
- A. 7cm, 12cm, 13cm
  - B. 5cm, 4cm, 6cm
  - C. 3cm, 4cm, 6cm
  - D. 12cm, 16cm, 20cm
27. A cylindrical container has an internal radius of 7cm and a height of 5cm. What is its capacity in litres? ( Take  $\pi = 22/7$  )
- A. 0.77
  - B. 7.7
  - C. 77
  - D. 770
28. A farmer used  $\frac{1}{2}$  of her land for planting maize,  $\frac{1}{5}$  for planting beans,  $\frac{1}{3}$  of the remainder for grazing and the rest for horticulture. If the farmer used 10 ha for grazing how much land does she have?
- A. 100ha
  - B.  $33 \frac{1}{3}$  ha
  - C. 30ha
  - D.  $11 \frac{1}{30}$
29. Which one of the triangles below has two of its side measuring 5cm and 7cm while one of its angles measures  $75^\circ$  ?
- A. Triangle KEF
  - B. Triangle JEF
  - C. Triangle HEF
  - D. Triangle GEF
30. In a certain company candidate G, K and L contested for set. The number of those who voted for K was 800, which was 0.25 of the total votes. Out of the remaining votes, L received 0.03 more than G. How many more votes than K did the winning candidate get?



- A. 72
- B. 364
- C. 436
- D. 448

31. The table below shows the second class train fare from station M to P through station N. The pupils paid fare as children.

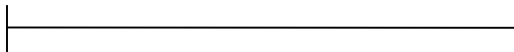
Three teachers accompanied 45 pupils in the train. The pupil fare only, from station M to P. Two of the teachers paid inclusive rate from station M to P. One teacher who alighted at station N paid for fare and beddings. How much money did they pay altogether?

- A. Sh. 25 195
- B. Sh. 28 020
- C. Sh. 28 325
- D. Sh. 50 520

32. The hire purchase price of a cupboard was 25% more than the marked price. Karana bought the cupboard on hire purchase terms. He deposited of 2 000 and eight equal installments of sh. 650. What was the marked price of the cupboard?

- A. Sh. 4, 160
- B. Sh. 5, 760
- C. Sh., 5 400
- D. Sh. 9 000

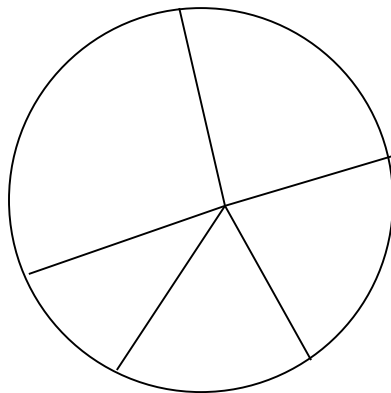
33. Construct a semi circle whose diameter EF is given below. Construct a line from E to meet the semi circle at G such that  $\angle FEG$  is  $30^\circ$ . Construct a line from F to meet the semi circle at H such that  $\angle EFH$  is  $20^\circ$ . Join points E to H to G and G to F.



What is the length of line GH?

- A. 3.0cm
- B. 3.9cm
- C. 5.3cm
- D. 5.7cm

34. A lorry has mass of 7.7 tonnes when loaded with 75 bags of rice. There are 33 bags each with mass of 85kg and the rest have mass of 45kg each. What is the mass of the lorry , in tonnes when empty?
- 3.005
  - 4.695
  - 4.850
  - 12.395
35. There were  $m$  men in a bus. The number of children in the bus was three time that of men but eleven more than that of women the total number of women, men and children in the bus was 45. Which one of the equations below can be used to find the number of men that were in a bus?
- $5m - 11 = 45$
  - $4m + 11 = 45$
  - $7m - 1 = 45$
  - $7m - 11 = 45$
36. Four workers take 10 hours to complete a certain job. How many more workers would be hored in order that they do the same job in 2 hours?
- 20
  - 24
  - 16
  - 1
37. The pie chart below shows how Kesenet spent her salary.



How much more did she spend on loan than on rent if she spent sh. 4, 000 on food?

- Sh. 5,000
- Sh.3,000

- C. Sh. 2,000  
D. Sh. 1,000
38. Which two of the following statements are true about triangles?
- (i) All angles are the same
  - (ii) Sum of interior angles is  $180^{\circ}$
  - (iii) One of the angle is  $90^{\circ}$
  - (iv) Sum of exterior angles is  $360^{\circ}$
- A. (i) and (ii)  
B. (i) and (iii)  
C. (ii) and (iv)  
D. (iii) and (iv)
39. There is 25% loss when an article is sold at sh.225. At what price should it be sold in order to make a profit of 5%
- A. Sh. 315.00  
B. Sh. 300.00  
C. Sh. 295.31  
D. Sh. 236.25
40. A player field measured 50m by 30m. The measurements of the sides of the field were later increased to 80m by 60m. What was the percentage increase of the area of the field?
- A. 4,800  
B. 3,300  
C. 900  
D. 220
41. How many more cubes are needed to fill the box below?
- A. 10  
B. 22  
C. 50  
D. 60
42. The figure below shows part of the journeys made by Chebet and Keya on the same road.
- Chebet travelled from town Y to town X at a constant speed. Keya travelled from town X to town Y. after covering 20km he rested for 30 minutes. He then continued at an average speed of 40km/h. Complete the graphs of the journeys?  
At what time did they meet?
- A. 9.00 a.m.  
B. 8.40 a.m  
C. 9. 10a.m  
D. 8.50a.m
43. Line PQ below is the base of the parallelogram PQRS. Complete the parallelogram in which, angle QPS =  $60^{\circ}$  and line PS = 4.5cm.



What is the height of the parallelogram in cm

- A. 3.9
- B. 5.4
- C. 6.0
- D. 9.1

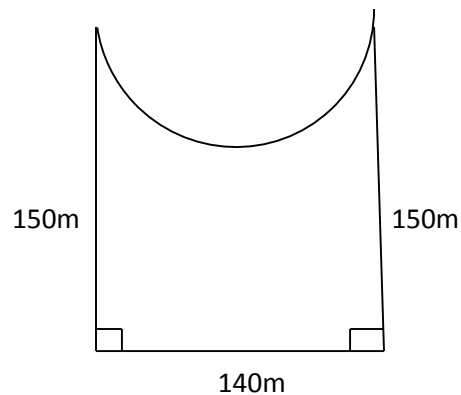
44. Karim paid sh. 950 for an item after getting a discount of 5%. What would have been the percentage discount, if Karim paid sh. 925 for the item?

- A.  $2\frac{12}{19}\%$
- B.  $7\frac{1}{2}\%$
- C.  $7\frac{17}{19}\%$
- D.  $8\frac{4}{47}\%$

45. A motorist driving at 60km/h was expected to arrive on time in town A, 200km away. After driving for one hour, the car got a puncture and it took 20 minutes to change the wheel. At what speed in km/h did he drive after repairing the puncture if he had to arrive at expected time?

- A. 105
- B. 100
- C. 70
- D.  $52\frac{1}{2}$

46. The diagram below represents a plot of land bounded by three straight edges and a semi circle. Three strands of wire are used to fence the plot.



What is the length, in metres, of the wire used?

(Take  $\pi = 22/7$ )

- A. 660
- B. 1,980
- C. 2,640
- D. 24,420

47. Which one of the following expressions is the simplest form of

$$\frac{7(2r+3)+4r-3}{2(r+1)+4r+7}$$

$$2(r+1)+4r+7$$

- A.  $\frac{6r+6}{2r+3}$
- B.  $\frac{6r+8}{2r+3}$
- C.  $\frac{9r}{3r+4}$

$$3r+4$$

- D.  $2\frac{1}{2}$

48. The table below shows the rate of sending letters and postcards through a post office in year 2004.

| Surface mail            |                   |                  |                         |                             |                                |
|-------------------------|-------------------|------------------|-------------------------|-----------------------------|--------------------------------|
| Type of article         | Weight step       | East Africa zone | The rest of Africa zone | Europe middle and near east | Australia America and far east |
| LETTERS Max. weight 2kg | Up to 20kg        | 40.00            | 45.00                   | 50kg                        | 60.00                          |
|                         | Over 20g to 100g  | 80.00            | 90.00                   | 100.00                      | 130.00                         |
|                         | Over 100g to 250g | 135.00           | 160.00                  | 180.00                      | 230.00                         |
|                         | Over 250g to 500g | 240.00           | 280.00                  | 320.00                      | 410.00                         |
|                         | Over 500g to 1kg  | 400.00           | 465.00                  | 530.00                      | 680.00                         |
|                         | Over 1kg to 2kg   | 650.00           | 760.00                  | 860.00                      | 1095.00                        |
| POST CARD               | Standard size     | 20.00            | 20.00                   | 25.00                       | 35.00                          |
|                         | large size        | 40.00            | 45.00                   | 50.00                       | 70.00                          |

Makena sent the following letters and postcards :

Two letters each weighing 21g; one of to Tanzania and another to Australia.

Four letters each weighing 280g one to Europe, two to America and one to Nigeria. Three large postcards; one to Australia one to Far East and one to America.

How much did she pay for postage altogether?

- A. Sh. 1,840.00
- B. Sh. 1,730.00
- C. Sh.1,820.00
- D. Sh. 1,430.00

49. The volume of an open rectangular tank is  $24.3\text{m}^3$ . The tank has a square base. The height of the tank is 2.7 metres. What is the surface area of the tank in square metres?

- A. 32.4
- B. 41.4
- C. 50.4
- D. 178.2

50. Janet borrowed some money at a simple interest of 12% p.a. After 18 months, she had paid a total interest of sh. 5,400. How much money had she paid altogether?

- A. Sh. 35,400
- B. Sh. 30,000
- C. Sh.24,600
- D. Sh. 7,900

